Person
+string PersonId [read-only]
+string FullName [read-only]
+string PhoneNumber [read-only]
#Person(string id, string name, string phone)
LibraryItem
+string CatalogNumber [read-only]
+string Title [read-only]
+decimal BaseLateFee [get, private set]
#decimal FeeMultiplier [get, set]
+decimal DailyLateFee [read-only]
+int LoanPeriodDays [get, protected set]
+bool IsWithdrawn [get, private set]
+bool IsOnLoan [get, private set]
#LibraryItem(string catalogNum, string title, decimal baseLateFee)
~SetBaseLateFee(decimal newFee)
~Withdraw()
~Restore()
~SetOnLoan(bool status)
1
involved in
0..*
Member
+int MaxItemsOnLoan [read-only]
+decimal DiscountPercentage [read-only]
-List<Loan> _loans
+IReadOnlyList<Loan> Loans [read-only]
#Member(string id, string name, string phone, int maxItems, decimal discount)
+Borrow(LibraryItem item, string loanId, DateTime borrowDate)
1
has
0..*
StaffMember
+DateTime HireDate [read-only]
+decimal Salary [get, private set]
#decimal ResponsibilityAllowance [get, set]
+decimal MonthlyPay [read-only]
#StaffMember(string id, string name, string phone, DateTime hireDate, decimal salary)
+GiveRaise(decimal percentage)
Book
+Book(string catalogNum, string title, decimal baseFee)
DVD
+DVD(string catalogNum, string title, decimal baseFee)
Magazine
+Magazine(string catalogNum, string title, decimal baseFee)
StudentMember
+StudentMember(string id, string name, string phone)
Loan
+string LoanId [read-only]
+DateTime BorrowDate [read-only]
+DateTime DueDate [read-only]
+LoanStatus Status [get, private set]
+DateTime ReturnDate [get, private set]
+Member Member [read-only]
+LibraryItem Item [read-only]
~Loan(string loanId, Member member, LibraryItem item, DateTime borrowDate)
~ReturnItem(DateTime returnDate)
~MarkLost()
+CalculateLateFee() : : decimal
*
status
1
«enumeration»
LoanStatus
Borrowed
Returned
Lost
PremiumMember
+int ReadingPoints [read-only]
+PremiumMember(string id, string name, string phone, decimal discount)
Librarian
+Librarian(string id, string name, string phone, DateTime hireDate, decimal salary)
+ProcessReturn(Loan loan, DateTime returnDate)
+MarkAsLost(Loan loan)
Shelver
+string Section [get, private set]
+Shelver(string id, string name, string phone, DateTime hireDate, decimal salary, string section)
+Reassign(string newSection)
HeadLibrarian
+HeadLibrarian(string id, string name, string phone, DateTime hireDate, decimal salary)
+AdjustBaseLateFee(LibraryItem item, decimal newFee)
+WithdrawItem(LibraryItem item)
+RestoreItem(LibraryItem item)